

Advanced TA Session

07/03/2026

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Probability Practice Sheet

Topics: Expectation, Variance, Covariance, Indicators, Joint
Distributions,
Independence, Occupancy, and Random Permutations

*First 20 pages contain one problem per page. Detailed solutions are
provided after the problem set.*

Problem

Problem 1: A company randomly assigns n jobs to n identical machines. Each job independently chooses a machine uniformly at random.

Let X be the number of machines that receive no jobs.

Find $\mathbf{E}[X]$.

Problem

Problem 2: A fair coin is tossed n times.

Let X be the number of adjacent pairs of tosses that are equal (that is, HH or TT).

Find $\mathbf{E}[X]$.

Problem

Problem 3: Two permutations of $\{1, 2, \dots, n\}$ are chosen independently and uniformly at random.

Let X be the number of positions in which the two permutations have the same value.

Find $\mathbf{E}[X]$.

Problem

Problem 4: m balls are thrown independently and uniformly at random into n bins.

Let X be the number of bins that receive at least one ball.

Find $\mathbf{E}[X]$.

Problem

Problem 5: A fair die is rolled repeatedly until a 6 appears.
Let X be the number of rolls required.
Compute $\mathbf{E}[X]$.

Problem

Problem 6: A permutation of $\{1, 2, \dots, n\}$ is chosen uniformly at random. Let X be the number of indices i such that $\pi(i) = i$. Find $\mathbf{E}[X]$.

Problem

[**Problem 7**] Let X take values -1 and 1 with probability $1/2$ each. Define

$$Y = X^2.$$

Compute $\text{Cov}(\mathbf{X}, \mathbf{Y})$.

Problem

Problem 8: A fair die is rolled n times.

Let X be the number of distinct faces that appear in the sequence of rolls.

Find $\mathbf{E}[X]$.

Problem

Problem 9: Let X be a random variable with variance $\text{Var}(X) = \sigma^2$.
Define

$$Y = 3X - 5.$$

Find $\text{Var}(Y)$.

Problem

[Problem 10] Two players independently toss a fair coin n times.
Let X be the number of times both players obtain the same outcome.
Find $\mathbf{E}[X]$.

Problem

Problem 11: A fair coin is tossed n times. A **run** is a maximal consecutive block of identical outcomes.

For example, the sequence

HHTTTHHH

has 4 runs: HH, TT, H, HH.

Let R denote the number of runs in the n tosses.

Find $\mathbf{E}[\mathbf{R}]$.

Problem

Problem 12: m balls are thrown independently and uniformly at random into n bins.

A **collision** is an unordered pair of balls that land in the same bin.

Let C be the total number of collisions.

Find $\mathbf{E}[C]$.

Problem

Problem 13: A permutation of $\{1, 2, \dots, n\}$ is chosen uniformly at random. Let X be the number of fixed points, that is,

$$X = \#\{i : \pi(i) = i\}.$$

Find $\text{Var}(X)$.

Problem

Problem 14: A permutation π of $\{1, 2, \dots, n\}$ is chosen uniformly at random. An **inversion** is a pair (i, j) with $1 \leq i < j \leq n$ such that

$$\pi(i) > \pi(j).$$

Let X be the total number of inversions.

Find $\mathbf{E}[\mathbf{X}]$.

Problem

[Problem 15] A permutation of $\{1, 2, \dots, n\}$ is chosen uniformly at random. For each $k \in \{1, 2, \dots, n-1\}$, consider the pair of values $(k, k+1)$. Let X be the number of such pairs that appear in adjacent positions in the permutation. Find $\mathbf{E}[X]$.

Problem

Problem 16: m balls are thrown independently and uniformly at random into n bins.

Let X be the number of bins that contain **exactly one** ball.

Find $\mathbf{E}[X]$.

Problem

Problem 17: Assume each of n people has a birthday independently and uniformly distributed over 365 days.

Let X be the number of unordered pairs of people having the same birthday.

Find $\mathbf{E}[X]$.

Problem

Problem 18: A permutation π of $\{1, 2, \dots, n\}$ is chosen uniformly at random. An index i is called a **record** if

$$\pi(i) > \max\{\pi(1), \pi(2), \dots, \pi(i-1)\}.$$

(So $i = 1$ is always a record.)

Let X be the number of records in the permutation.

Find $\mathbf{E}[X]$.

Problem

Problem 19: A fair coin is tossed $n + 1$ times. Let

$H_1 =$ number of heads in tosses $1, 2, \dots, n$,

and

$H_2 =$ number of heads in tosses $2, 3, \dots, n + 1$.

Find $\text{Cov}(\mathbf{H}_1, \mathbf{H}_2)$.

Problem

Problem 20: A fair coin is tossed n times.

Let X be the number of adjacent equal pairs, that is, the number of indices $i \in \{1, \dots, n-1\}$ for which toss i and toss $i+1$ are the same.

Find $\text{Var}(\mathbf{X})$.

Detailed Stepwise Solutions

Detailed Solution

Solution to Problem 1: Let

$$X_i = \begin{cases} 1, & \text{if machine } i \text{ receives no job,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{i=1}^n X_i.$$

Step 1: Use linearity of expectation

$$E[X] = \sum_{i=1}^n E[X_i].$$

Step 2: Compute $E[X_i]$

Since X_i is an indicator,

$$E[X_i] = P(\text{machine } i \text{ receives no job}).$$

A single job avoids machine i with probability

$$1 - \frac{1}{n}.$$

Since all n jobs choose independently,

$$P(\text{machine } i \text{ receives no job}) = \left(1 - \frac{1}{n}\right)^n.$$

Hence

$$E[X_i] = \left(1 - \frac{1}{n}\right)^n.$$

Step 3: Final answer

$$E[X] = n \left(1 - \frac{1}{n}\right)^n.$$

$$E[X] = n \left(1 - \frac{1}{n}\right)^n$$

Detailed Solution

Solution to Problem 2: For $i = 1, 2, \dots, n-1$, define

$$X_i = \begin{cases} 1, & \text{if tosses } i \text{ and } i+1 \text{ are equal,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{i=1}^{n-1} X_i.$$

Step 1: Use linearity of expectation

$$E[X] = \sum_{i=1}^{n-1} E[X_i].$$

Step 2: Compute $E[X_i]$

Two consecutive fair tosses can be

HH, HT, TH, TT.

Out of these four outcomes, two are equal-pairs: HH and TT.

So

$$P(X_i = 1) = \frac{2}{4} = \frac{1}{2}.$$

Thus

$$E[X_i] = \frac{1}{2}.$$

Step 3: Final answer

$$E[X] = (n-1) \cdot \frac{1}{2} = \frac{n-1}{2}.$$

$$E[X] = \frac{n-1}{2}$$

Detailed Solution

Solution to Problem 3: Let

$$X_i = \begin{cases} 1, & \text{if the two permutations agree at position } i, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{i=1}^n X_i.$$

Step 1: Use linearity of expectation

$$E[X] = \sum_{i=1}^n E[X_i].$$

Step 2: Compute $E[X_i]$

Fix the first permutation. In the second permutation, the entry at position i is equally likely to be any of the n symbols. Therefore the probability that it matches the first permutation at that position is

$$\frac{1}{n}.$$

Hence

$$E[X_i] = \frac{1}{n}.$$

Step 3: Final answer

$$E[X] = n \cdot \frac{1}{n} = 1.$$

$$\boxed{E[X] = 1}$$

Detailed Solution

Solution to Problem 4: Define

$$X_i = \begin{cases} 1, & \text{if bin } i \text{ gets at least one ball,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{i=1}^n X_i.$$

Step 1: Use linearity of expectation

$$E[X] = \sum_{i=1}^n E[X_i].$$

Step 2: Compute $E[X_i]$

For a fixed bin i , the probability that a single ball avoids it is

$$1 - \frac{1}{n}.$$

Hence the probability that all m balls avoid it is

$$\left(1 - \frac{1}{n}\right)^m.$$

Therefore

$$P(\text{bin } i \text{ is occupied}) = 1 - \left(1 - \frac{1}{n}\right)^m.$$

So

$$E[X_i] = 1 - \left(1 - \frac{1}{n}\right)^m.$$

Step 3: Final answer

$$E[X] = n \left(1 - \left(1 - \frac{1}{n}\right)^m\right).$$

$$\boxed{E[X] = n \left(1 - \left(1 - \frac{1}{n}\right)^m\right)}$$

Detailed Solution

Solution to Problem 5: Let

$$I_k = \begin{cases} 1, & \text{if the } k\text{th roll actually occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Then the number of rolls until the first 6 is

$$X = \sum_{k=1}^{\infty} I_k.$$

Step 1: Take expectation

$$E[X] = \sum_{k=1}^{\infty} E[I_k].$$

Step 2: Compute $E[I_k]$

The k th roll occurs iff the first $k - 1$ rolls are not 6.

A single roll is not 6 with probability $\frac{5}{6}$, so

$$P(I_k = 1) = \left(\frac{5}{6}\right)^{k-1}.$$

Hence

$$E[I_k] = \left(\frac{5}{6}\right)^{k-1}.$$

Step 3: Sum the geometric series

$$E[X] = \sum_{k=1}^{\infty} \left(\frac{5}{6}\right)^{k-1} = \frac{1}{1 - \frac{5}{6}} = 6.$$

$$\boxed{E[X] = 6}$$

Detailed Solution

Solution to Problem 6: Let

$$X_i = \begin{cases} 1, & \text{if } \pi(i) = i, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{i=1}^n X_i.$$

Step 1: Use linearity of expectation

$$E[X] = \sum_{i=1}^n E[X_i].$$

Step 2: Compute $E[X_i]$

In a uniformly random permutation, any of the n values is equally likely to appear at position i . Therefore

$$P(\pi(i) = i) = \frac{1}{n}.$$

So

$$E[X_i] = \frac{1}{n}.$$

Step 3: Final answer

$$E[X] = n \cdot \frac{1}{n} = 1.$$

$$\boxed{E[X] = 1}$$

Detailed Solution

Solution to Problem 7: Since $X \in \{-1, 1\}$, we have

$$Y = X^2 = 1$$

always.

Step 1: Compute $E[X]$

$$E[X] = (-1) \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = 0.$$

Step 2: Compute $E[Y]$

$$E[Y] = 1.$$

Step 3: Compute $E[XY]$

Since $Y = 1$,

$$XY = X \cdot 1 = X.$$

Hence

$$E[XY] = E[X] = 0.$$

Step 4: Apply covariance formula

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

So

$$\text{Cov}(X, Y) = 0 - (0)(1) = 0.$$

$$\boxed{\text{Cov}(X, Y) = 0}$$

Detailed Solution

Solution to Problem 8: For each face $i \in \{1, 2, 3, 4, 5, 6\}$, define

$$X_i = \begin{cases} 1, & \text{if face } i \text{ appears at least once,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{i=1}^6 X_i.$$

Step 1: Use linearity of expectation

$$E[X] = \sum_{i=1}^6 E[X_i].$$

Step 2: Compute $E[X_i]$

A fixed face i does not appear on a single roll with probability $\frac{5}{6}$. Thus it does not appear in n rolls with probability

$$\left(\frac{5}{6}\right)^n.$$

Therefore

$$P(X_i = 1) = 1 - \left(\frac{5}{6}\right)^n.$$

So

$$E[X_i] = 1 - \left(\frac{5}{6}\right)^n.$$

Step 3: Final answer

$$E[X] = 6 \left(1 - \left(\frac{5}{6}\right)^n\right).$$

$$E[X] = 6 \left(1 - \left(\frac{5}{6}\right)^n\right)$$

Detailed Solution

Solution to Problem 9: We use the standard variance transformation rule:

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

Here

$$Y = 3X - 5.$$

So

$$\text{Var}(Y) = \text{Var}(3X - 5) = 3^2 \text{Var}(X) = 9\sigma^2.$$

$$\text{Var}(Y) = 9\sigma^2$$

Detailed Solution

Solution to Problem 10: For each toss $i = 1, 2, \dots, n$, define

$$X_i = \begin{cases} 1, & \text{if the two players have the same outcome on toss } i, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{i=1}^n X_i.$$

Step 1: Use linearity of expectation

$$E[X] = \sum_{i=1}^n E[X_i].$$

Step 2: Compute $E[X_i]$

For one toss, the outcome-pairs are

HH, HT, TH, TT,

all equally likely. The matching outcomes are HH and TT, so

$$P(X_i = 1) = \frac{2}{4} = \frac{1}{2}.$$

Thus

$$E[X_i] = \frac{1}{2}.$$

Step 3: Final answer

$$E[X] = n \cdot \frac{1}{2} = \frac{n}{2}.$$

$$\boxed{E[X] = \frac{n}{2}}$$

Detailed Solution

Solution to Problem 11: Let the tosses be $T_1, T_2, \dots, T_n$. A new run starts at position 1, and thereafter a new run starts whenever the current toss differs from the previous toss.

Define, for $i = 1, 2, \dots, n-1$,

$$I_i = \begin{cases} 1, & \text{if } T_i \neq T_{i+1}, \\ 0, & \text{otherwise.} \end{cases}$$

Then the total number of runs is

$$R = 1 + \sum_{i=1}^{n-1} I_i.$$

Step 1: Take expectation

$$E[R] = 1 + \sum_{i=1}^{n-1} E[I_i].$$

Step 2: Compute $E[I_i]$

For two consecutive fair tosses, the outcomes differ in exactly two cases:

HT, TH.

Hence

$$P(I_i = 1) = \frac{1}{2}.$$

So

$$E[I_i] = \frac{1}{2}.$$

Step 3: Final answer

$$E[R] = 1 + (n-1) \cdot \frac{1}{2} = 1 + \frac{n-1}{2} = \frac{n+1}{2}.$$

$$E[R] = \frac{n+1}{2}$$

Detailed Solution

Solution to Problem 12: Label the balls $1, 2, \dots, m$. For each unordered pair $\{a, b\}$ with $1 \leq a < b \leq m$, define

$$I_{ab} = \begin{cases} 1, & \text{if balls } a \text{ and } b \text{ land in the same bin,} \\ 0, & \text{otherwise.} \end{cases}$$

Then the number of collisions is

$$C = \sum_{1 \leq a < b \leq m} I_{ab}.$$

Step 1: Use linearity of expectation

$$E[C] = \sum_{1 \leq a < b \leq m} E[I_{ab}].$$

Step 2: Compute $E[I_{ab}]$

Fix balls a and b . Once ball a chooses a bin, ball b must choose that same bin to create a collision. Since each ball chooses uniformly among n bins,

$$P(I_{ab} = 1) = \frac{1}{n}.$$

Thus

$$E[I_{ab}] = \frac{1}{n}.$$

Step 3: Count the number of unordered pairs

There are

$$\binom{m}{2}$$

unordered pairs of balls.

Step 4: Final answer

$$E[C] = \binom{m}{2} \cdot \frac{1}{n}.$$

Equivalently,

$$E[C] = \frac{m(m-1)}{2n}.$$

$$E[C] = \frac{\binom{m}{2}}{n} = \frac{m(m-1)}{2n}$$

Detailed Solution

Solution to Problem 13: Let

$$X_i = \begin{cases} 1, & \text{if } \pi(i) = i, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{i=1}^n X_i.$$

We want

$$\text{Var}(X).$$

Step 1: Use the variance expansion

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j).$$

Step 2: Compute $\text{Var}(X_i)$

We know

$$P(X_i = 1) = \frac{1}{n}.$$

So X_i is Bernoulli with parameter $\frac{1}{n}$, hence

$$\text{Var}(X_i) = \frac{1}{n} \left(1 - \frac{1}{n} \right).$$

Therefore

$$\sum_{i=1}^n \text{Var}(X_i) = n \cdot \frac{1}{n} \left(1 - \frac{1}{n} \right) = 1 - \frac{1}{n}.$$

Step 3: Compute $\text{Cov}(X_i, X_j)$ for $i \neq j$

We have

$$\text{Cov}(X_i, X_j) = E[X_i X_j] - E[X_i]E[X_j].$$

Now $X_i X_j = 1$ iff both i and j are fixed points. The number of permutations fixing both i and j is $(n-2)!$, so

$$P(X_i = 1, X_j = 1) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}.$$

Also,

$$E[X_i]E[X_j] = \frac{1}{n} \cdot \frac{1}{n} = \frac{1}{n^2}.$$

Thus

$$\text{Cov}(X_i, X_j) = \frac{1}{n(n-1)} - \frac{1}{n^2} = \frac{1}{n^2(n-1)}.$$

Step 4: Sum the covariance terms

There are $\binom{n}{2}$ pairs (i, j) with $i < j$, so

$$2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j) = 2 \binom{n}{2} \cdot \frac{1}{n^2(n-1)}.$$

Since

$$2 \binom{n}{2} = n(n-1),$$

this becomes

$$\frac{n(n-1)}{n^2(n-1)} = \frac{1}{n}.$$

Step 5: Final answer

$$\text{Var}(X) = \left(1 - \frac{1}{n}\right) + \frac{1}{n} = 1.$$

$\text{Var}(X) = 1$

Detailed Solution

Solution to Problem 14: For each pair (i, j) with $1 \leq i < j \leq n$, define

$$I_{ij} = \begin{cases} 1, & \text{if } \pi(i) > \pi(j), \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{1 \leq i < j \leq n} I_{ij}.$$

Step 1: Use linearity of expectation

$$E[X] = \sum_{1 \leq i < j \leq n} E[I_{ij}].$$

Step 2: Compute $E[I_{ij}]$

For a fixed pair $i < j$, the two values $\pi(i)$ and $\pi(j)$ are equally likely to appear in either relative order. Thus

$$P(\pi(i) > \pi(j)) = \frac{1}{2}.$$

Hence

$$E[I_{ij}] = \frac{1}{2}.$$

Step 3: Count the pairs

There are

$$\binom{n}{2}$$

choices of $i < j$.

Step 4: Final answer

$$E[X] = \binom{n}{2} \cdot \frac{1}{2} = \frac{n(n-1)}{4}.$$

$$E[X] = \frac{n(n-1)}{4}$$

Detailed Solution

Solution to Problem 15 For each $k = 1, 2, \dots, n-1$, define

$$I_k = \begin{cases} 1, & \text{if the values } k \text{ and } k+1 \text{ are adjacent in the permutation,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{k=1}^{n-1} I_k.$$

Step 1: Use linearity of expectation

$$E[X] = \sum_{k=1}^{n-1} E[I_k].$$

Step 2: Compute $E[I_k]$

Fix k . We count permutations where k and $k+1$ appear adjacent.

Treat $(k, k+1)$ as a single block. Then we have $(n-1)$ objects to permute:

$$\{\text{block}, \text{other } n-2 \text{ values}\}.$$

These can be arranged in

$$(n-1)!$$

ways. Inside the block, the two values can be ordered as

$$(k, k+1) \quad \text{or} \quad (k+1, k),$$

so there are 2 internal orders.

Hence the number of favorable permutations is

$$2(n-1)!.$$

Since the total number of permutations is $n!$,

$$P(I_k = 1) = \frac{2(n-1)!}{n!} = \frac{2}{n}.$$

Therefore

$$E[I_k] = \frac{2}{n}.$$

Step 3: Final answer

$$E[X] = (n-1) \cdot \frac{2}{n} = \frac{2(n-1)}{n}.$$

$$E[X] = \frac{2(n-1)}{n}$$

Detailed Solution

Solution to Problem 16: For each bin $i = 1, 2, \dots, n$, define

$$X_i = \begin{cases} 1, & \text{if bin } i \text{ contains exactly one ball,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{i=1}^n X_i.$$

Step 1: Use linearity of expectation

$$E[X] = \sum_{i=1}^n E[X_i].$$

Step 2: Compute $E[X_i]$

We want the probability that bin i receives exactly one ball.

Choose which one of the m balls lands in bin i : there are m choices.

For that chosen ball, the probability of landing in bin i is

$$\frac{1}{n}.$$

For each of the other $m-1$ balls, the probability of avoiding bin i is

$$1 - \frac{1}{n}.$$

So

$$P(X_i = 1) = m \cdot \frac{1}{n} \left(1 - \frac{1}{n}\right)^{m-1}.$$

Hence

$$E[X_i] = m \cdot \frac{1}{n} \left(1 - \frac{1}{n}\right)^{m-1}.$$

Step 3: Final answer

$$E[X] = n \cdot m \cdot \frac{1}{n} \left(1 - \frac{1}{n}\right)^{m-1} = m \left(1 - \frac{1}{n}\right)^{m-1}.$$

$$E[X] = m \left(1 - \frac{1}{n}\right)^{m-1}$$

Detailed Solution

Solution to Problem 17: For each unordered pair of people $\{i, j\}$ with $1 \leq i < j \leq n$, define

$$I_{ij} = \begin{cases} 1, & \text{if persons } i \text{ and } j \text{ have the same birthday,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{1 \leq i < j \leq n} I_{ij}.$$

Step 1: Use linearity of expectation

$$E[X] = \sum_{1 \leq i < j \leq n} E[I_{ij}].$$

Step 2: Compute $E[I_{ij}]$

For a fixed pair (i, j) , once person i 's birthday is chosen, person j must match it. Since birthdays are uniformly distributed over 365 days,

$$P(I_{ij} = 1) = \frac{1}{365}.$$

Hence

$$E[I_{ij}] = \frac{1}{365}.$$

Step 3: Count the number of unordered pairs

There are

$$\binom{n}{2}$$

pairs.

Step 4: Final answer

$$E[X] = \binom{n}{2} \cdot \frac{1}{365} = \frac{n(n-1)}{2 \cdot 365}.$$

$$E[X] = \frac{\binom{n}{2}}{365} = \frac{n(n-1)}{730}$$

Detailed Solution

Solution to Problem 18: For each position $i = 1, 2, \dots, n$, define

$$I_i = \begin{cases} 1, & \text{if position } i \text{ is a record,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{i=1}^n I_i.$$

Step 1: Use linearity of expectation

$$E[X] = \sum_{i=1}^n E[I_i].$$

Step 2: Compute $E[I_i]$

Position i is a record iff the value at position i is the largest among the first i entries

$$\pi(1), \pi(2), \dots, \pi(i).$$

By symmetry, each of these i positions is equally likely to contain the largest value among the first i entries. Therefore

$$P(I_i = 1) = \frac{1}{i}.$$

Hence

$$E[I_i] = \frac{1}{i}.$$

Step 3: Final answer

$$E[X] = \sum_{i=1}^n \frac{1}{i}.$$

Thus

$$E[X] = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$$

which is the n th harmonic number.

Detailed Solution

Solution to Problem 19: Let the $(n+1)$ coin tosses be represented by indicators

$$Y_1, Y_2, \dots, Y_{n+1},$$

where

$$Y_i = \begin{cases} 1, & \text{if toss } i \text{ is Head,} \\ 0, & \text{if toss } i \text{ is Tail.} \end{cases}$$

Then

$$H_1 = Y_1 + Y_2 + \dots + Y_n, \quad H_2 = Y_2 + Y_3 + \dots + Y_{n+1}.$$

Define

$$S = Y_2 + Y_3 + \cdots + Y_n.$$

Then

$$H_1 = Y_1 + S, \quad H_2 = S + Y_{n+1}.$$

Step 1: Expand the covariance

$$\text{Cov}(H_1, H_2) = \text{Cov}(Y_1 + S, S + Y_{n+1}).$$

Using bilinearity of covariance,

$$\text{Cov}(H_1, H_2) = \text{Cov}(Y_1, S) + \text{Cov}(Y_1, Y_{n+1}) + \text{Cov}(S, S) + \text{Cov}(S, Y_{n+1}).$$

Step 2: Use independence

Since the coin tosses are independent, Y_1 is independent of S , Y_1 is independent of Y_{n+1} , and S is independent of Y_{n+1} . Therefore

$$\text{Cov}(Y_1, S) = 0, \quad \text{Cov}(Y_1, Y_{n+1}) = 0, \quad \text{Cov}(S, Y_{n+1}) = 0.$$

Hence

$$\text{Cov}(H_1, H_2) = \text{Cov}(S, S) = \text{Var}(S).$$

Step 3: Compute $\text{Var}(S)$

The sum S contains $(n-1)$ independent Bernoulli(1/2) variables. Each has variance

$$\text{Var}(Y_i) = \frac{1}{4}.$$

So

$$\text{Var}(S) = (n-1) \cdot \frac{1}{4}.$$

Step 4: Final answer

$$\boxed{\text{Cov}(H_1, H_2) = \frac{n-1}{4}}$$

Detailed Solution

Solution to Problem 20: For $i = 1, 2, \dots, n-1$, define

$$I_i = \begin{cases} 1, & \text{if tosses } i \text{ and } i+1 \text{ are equal,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$X = \sum_{i=1}^{n-1} I_i.$$

We want

$$\text{Var}(X).$$

Step 1: Use the variance expansion

$$\text{Var}(X) = \sum_{i=1}^{n-1} \text{Var}(I_i) + 2 \sum_{1 \leq i < j \leq n-1} \text{Cov}(I_i, I_j).$$

Step 2: Compute $\text{Var}(I_i)$

We know

$$P(I_i = 1) = \frac{1}{2}.$$

So I_i is Bernoulli(1/2) and

$$\text{Var}(I_i) = \frac{1}{2} \left(1 - \frac{1}{2} \right) = \frac{1}{4}.$$

Hence

$$\sum_{i=1}^{n-1} \text{Var}(I_i) = (n-1) \cdot \frac{1}{4}.$$

Step 3: Compute covariance terms

Case 1: $|i - j| \geq 2$

Then I_i and I_j depend on disjoint sets of coin tosses, so they are independent. Thus

$$\text{Cov}(I_i, I_j) = 0.$$

Case 2: $j = i + 1$

We compute

$$\text{Cov}(I_i, I_{i+1}) = E[I_i I_{i+1}] - E[I_i]E[I_{i+1}].$$

Now $I_i I_{i+1} = 1$ iff tosses $i, i+1, i+2$ are all the same. The only possibilities are

HHH or TTT.

Each has probability $\frac{1}{8}$, so

$$P(I_i = 1, I_{i+1} = 1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.$$

Also

$$E[I_i]E[I_{i+1}] = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Hence

$$\text{Cov}(I_i, I_{i+1}) = \frac{1}{4} - \frac{1}{4} = 0.$$

Thus *all* covariance terms are zero.

Step 4: Final answer

$$\text{Var}(X) = (n-1) \cdot \frac{1}{4}.$$

$$\boxed{\text{Var}(X) = \frac{n-1}{4}}$$